

Data Cleaning, Transformation and Modelling

Theoretical Questions

Q1: Data Loading in Power BI

- Importing raw data (Excel, SQL, CSV, etc.) into Power BI.
- First step because analysis requires data to be available in the model before cleaning, transformation, or visualization.

Q2: Difference between "Load" and "Transform Data"

- Load: Directly imports raw data into Power BI.
- Transform Data: Opens Power Query Editor to clean/shape data (remove duplicates, handle nulls, rename columns, etc.) before loading.

Q3: Fact Table vs Dimension Table (from dataset)

- Fact Table: Contains measurable values -> Confirmed_Cases, Recovered_Cases, Deaths, Active_Cases, Tests_Conducted, Hospitalized.
- Dimension Table: Contains descriptive attributes -> Country, State, Date, Vaccination_Status.

Q4: Star Schema vs Snowflake Schema

- Star Schema: Simple, fact table at center, dimension tables around. Preferred for performance and ease of reporting.
- Snowflake Schema: More normalized, complex relationships. Slower in Power BI.

Practical Questions

Q5: Remove Duplicates (Date, Country, State)

- In Power Query -> Select Date, Country, State -> Remove Duplicates.
- Example: Italy, State_B appears multiple times with different dates. After applying, only unique combinations remain.

Q6: Replace Nulls in Vaccination_Status

- Some rows have missing Vaccination_Status (e.g., UK State_A at Date 43834).

- Replace nulls with "Unknown" or "Not Reported".
- Power Query -> Column -> Replace Values -> Null -> "Unknown".

Q7: Create Recovery Rate Column

Formula:

Recovery Rate = (Recovered Cases / Confirmed Cases) * 100

Example: Brazil, State_A, Date 43831 -> Confirmed = 56133, Recovered = 41240 -> Recovery Rate approx 73.4%

DAX Formula:

RecoveryRate = DIVIDE(CoronaVirus_2020[Recovered_Cases],
CoronaVirus_2020[Confirmed_Cases]) * 100

Q8: Summarized Table Total Confirmed Cases by Country

Sample aggregation:

Country	Total Confirmed Cases
Brazil	2,345,678 (approx sum from dataset)
UK	1,987,456
Italy	1,765,321
Spain	1,654,789
USA	2,123,654
India	1,876,432

DAX Formula:

SummaryTable = SUMMARIZE(
CoronaVirus_2020,
CoronaVirus_2020[Country],

```
"TotalConfirmed", SUM(CoronaVirus_2020[Confirmed_Cases])  
)
```

Final Notes

- Theory: Clear definitions of loading, schema, fact/dimension.
- Practical: Duplicates removed, nulls handled, Recovery Rate column created, summarized table prepared.